

In-class Chapter 5 and 6: Variables and Data Types

Chapter 5:

1. Assume the following Ada program was compiled and executed.

```
procedure Main is
  X: Integer;
  procedure Sub1 is
    begin -- of Sub1
      Put(X);
    end;
  procedure Sub2 is
    X: Integer;
    begin -- of Sub2
      X := 10;
      Sub1;
    end;
  begin -- of Main
    X := 5;
    Sub2
  end;
```

a) Using static scoping rules, what is the value of X is printed in procedure Sub1?

5

b) Under dynamic-scoping rules, what value of X is printed in procedure Sub1?

10

Chapter 6:

2. There are five different types of arrays discussed in class: static, fixed stack-dynamic, fixed heap-dynamic, and heap-dynamic. Identify the categories of arrays at positions 1, 2, 3, and 4.

```
int foo(void) {
  int z[2];           _____→ 1
  static int a[3] = {3, 6, 9} _____→ 2
  int* b = new int[6]; _____→ 3
  vector<int> c;      _____→ 4
}
```

1) Int Z[2] → fixed stack dynamic array

2) Static int a[3] → static array

3) Int* b = new int[6] → fixed heap dynamic array

4) Vector<int> c → heap dynamic array

3. Given all the languages that you know, produce a list of data types in these languages.

Int, string, list, tuple, Boolean, float, long,

4. What are the advantages of user-defined enumeration types?

Improved readability and maintainability for example descriptive names for a set of values. Another example is type safety where users can restrict the variables to only accept values from the defined set of enum members.

5. List two design issues for arrays.

Slicing - what type of array slicing is allowed and how are they accessed

Memory management - potential memory waste

6. What is the structure of an associative array?

It contains key value pairs, unique keys and dynamic sizing

7. List one aspect that record and union are the same and one aspect that they are different.

Similarity is they can both be used for error representation

Differences are scope between record and union

8. Are the tuples of Python mutable?

no

9. Java is a static-typed language. What does "static-typed language" mean?

It means a programming language checks the data type of variables at compile time.

10. Compared with C, Java is a strong-typed language. What does "strong-typed language" mean?

Strongly typed languages prevent mixing incompatible types and catches type errors at compile time.

11. What is a compatible type?

A compatible type is programming language that allows it to be used in place of another type without causing errors or violating the languages rules.

12. How do you evaluate a programming language by its data types?

There are a few ways to evaluate this some are support for abstraction and expressiveness of the language does it offer a variety of data types etc.